

Darin Davis Johnson

(+91) 9380321413 | darinjohn23@gmail.com | LinkedIn - darin-davis-johnson | GitHub - DarinDJ | darindj.github.io/MyWebsite

PROFESSIONAL EXPERIENCE

KOREA INSTITUTE OF SCIENCE AND TECHNOLOGY (KIST)

Seoul, South Korea

Student Researcher (UST-KIST M.S. Program)

September 2026 – Present

- Developing research software in Python to process, analyze and visualize mouse neural-recording and behavioral data, turning experimental data into reproducible analysis pipelines.
- Applying AI and machine-learning methods to mouse neuroscience data as part of the M.S. in AI and Robotics.

Research Intern

May 2026 – August 2026

- Built an end-to-end Python pipeline (PSD, artifact masking, Hilbert-envelope burst detection) to analyze LFP/EEG recordings from the mPFC and basolateral amygdala in a mouse threat-escape experiment.
- Detected and visually validated theta, beta and gamma bursts, then compared burst features across four 60-second experimental conditions using statistical analysis (SciPy, statsmodels).
- Aligned neural bursts with tracked mouse and spider-robot behavior and analyzed beta-gamma-theta cross-frequency coupling to link brain dynamics to threat-driven behavior.

MICROHARD SERVICES PVT LTD.

Bengaluru, India

Software Engineering Intern

May 2025 - June 2025

- Built backend logic in Python and SQL for an internal Asset Tracking Application, enabling real-time updates across 100+ items.
- Reduced manual tracking overhead by ~40% while collaborating with the development team on functional components and design.

HINDUSTAN AERONAUTICS LIMITED (HAL)

Bengaluru, India

Software Engineering Intern

April 2024 - May 2024

- Developed a mission-critical Java application (Eclipse IDE) for helicopter pilots, delivering real-time data visualization and seamless system communication.
- Engineered a secure LAN-based interface to receive and display real-time operational data from an external system, adhering to strict security protocols.
- Optimized data handling and UI responsiveness, reducing latency by 30%.
- Delivered 80% of the assigned scope within one month of a two-month timeline, accelerating completion by 50%.

EDUCATION

UNIVERSITY OF SCIENCE AND TECHNOLOGY (UST) – KIST SCHOOL

Seoul, South Korea

Master of Science in AI and Robotics

September 2026 - Present

- Research focus: applied AI for neuroscience and research software engineering.

CHRIST UNIVERSITY

Bengaluru, India

Bachelor of Technology in Computer Science and Engineering with Data Science

August 2022 - April 2026

- Average GPA: 3.7/4 | Relevant coursework: Data Structures, Algorithms, AI, ML, Databases, SQL.

CERTIFICATIONS AND PROJECTS

BANK CUSTOMER CHURN PREDICTION

March 2024

- Compared six classifiers (logistic regression, SVM, KNN, decision tree, random forest, gradient boosting) on 10,000 customer records to predict churn, handling class imbalance with resampling; random forest performed best.
- Evaluated with accuracy, precision, recall and F1, and wrapped the model in a Tkinter app for interactive predictions.

FACIAL RECOGNITION & EMOTION DETECTION

September 2024

- Built a CNN-based deep-learning system (OpenCV, TensorFlow, Keras) that detects faces and classifies emotions in real time.

KERALEEYAM ASSOCIATION WEBSITE

February 2025

- Built and deployed a full-stack website (React.js, TypeScript, Supabase) for 50+ association members; live at keralaleeyam.vercel.app.
- Automated database integration for dynamic content and user interactions, cutting manual effort by 60%; collaborated in a team of 5 with 100% on-time deployment.

BRITISH AIRWAYS DATA SCIENCE VIRTUAL EXPERIENCE – FORAGE

July 2023

- Scraped and analyzed airline customer data (Python, BeautifulSoup) and built a logistic-regression model forecasting buying behavior with ~90% accuracy; visualized booking trends with Matplotlib.

NoSQL and DBaaS 101 - IBM

October 2023

INTRODUCTION TO CYBERSECURITY - CISCO

June 2023

SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL

ML & Data: scikit-learn, TensorFlow, Keras, OpenCV, NumPy, pandas, SciPy, statsmodels, Matplotlib, Power BI

Neuro & Signals: MNE-Python, power spectral density, Hilbert transform, burst detection, cross-frequency coupling

Tools: Git, REST APIs, React.js, Supabase, MySQL, Jupyter, Raspberry Pi